

valores y vectores propios

$$Au = \lambda u$$

$$(A - I\lambda)u = 0$$

$$|A - I\lambda| = 0$$

polinomio de grado n

$$\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$$

$$u_1, u_2, u_3, \dots, u_n$$

$$E(X) = \begin{pmatrix} E(x_1) \\ E(x_2) \\ \vdots \\ E(x_p) \end{pmatrix} \quad \text{vector}$$

$$Y = AX + B \quad E(Y) = AE(X) + B$$

A, B matrices de constantes

$$V(X) = E(X - E(X))^2 = E(X^2) - E(X)^2$$

$$Cov(X, Y) = E(X - E(X))(Y - E(Y))$$

$$\Sigma_X = Cov(X) = E[(X - E(X))(X - E(X))^T]$$

$$= \begin{pmatrix} E(x_1 - E(x_1))(x_1 - E(x_1)) & E(x_1 - E(x_1))(x_2 - E(x_2)) & \dots & E(x_1 - E(x_1))(x_n - E(x_n)) \\ E(x_2 - E(x_2))(x_1 - E(x_1)) & E(x_2 - E(x_2))(x_2 - E(x_2)) & \dots & E(x_2 - E(x_2))(x_n - E(x_n)) \\ \vdots & \vdots & \ddots & \vdots \\ E(x_n - E(x_n))(x_1 - E(x_1)) & E(x_n - E(x_n))(x_2 - E(x_2)) & \dots & E(x_n - E(x_n))(x_n - E(x_n)) \end{pmatrix}$$

$$1) \bar{\Sigma} = \bar{\Sigma}^T$$

2) $\bar{\Sigma}$ semidefinida positiva

3) $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$

$$Y = AX + b$$

$$\Sigma_Y = A \Sigma_X A^T$$

Componentes principales

* Reducir dimensión

$$\rightarrow y_j = a_{j1}x_1 + a_{j2}x_2 + \dots + a_{jp}x_p \quad j=1, 2, \dots, p$$

obtener p nuevas variables no correlacionadas
($Cov(x_i, x_j) = 0$)

$$V(y_1) \geq V(y_2) \geq V(y_3) \geq \dots \geq V(y_p)$$

No correlación

$$a_i^T a_j = 0$$

$$a_i^T a_i = 1$$

a_i 's ortogonales

$$y_1 = a_1^T X \quad \text{1er componente}$$

$$V(y_1) = a_1^T V(X) a_1 = a_1^T \Sigma_X a_1 \leftarrow \text{maximizar}$$

$$\text{sujeta a } a_1^T a_1 = 1$$

Multiplicadores de Lagrange

$$L(a_1) = a_1^T \Sigma_X a_1 - \lambda (a_1^T a_1 - 1)$$

$$\frac{d}{dz} A z = A \quad \leftarrow \text{matriz const.}$$

$$\frac{d}{dz} z^T z = 2z$$

$$\frac{d}{da_1} L(a_1) = 2 \Sigma_X a_1 - \lambda 2 a_1$$

$$\Sigma_X a_1 - \lambda a_1 = 0$$

a_1 es el vector propio de Σ_X

$$\Sigma_X a_1 = \lambda a_1$$

λ_1 el valor más grande

$$y_1 = a_1^T X \quad V(y_1) = a_1^T \Sigma_X a_1$$

$$= a_1^T \lambda a_1 = \lambda a_1^T a_1 = \lambda$$

$\Rightarrow \lambda_1$ como el valor propio más grande

Calculamos y_2

* $V(y_2)$ la 2ª más grande

$$Cov(y_1, y_2) = 0$$

$$\Rightarrow V(y_2) = V(a_2^T X) = a_2^T \Sigma_X a_2$$

$$Cov(y_1, y_2) = E(y_1 - E(y_1))(y_2 - E(y_2))^T$$

$$= E(a_1^T X - a_1^T E(X))(a_2^T X - a_2^T E(X))^T$$

$$= a_1^T \Sigma_X a_2$$

$$0 = Cov(y_2, y_1) = a_2^T \Sigma_X a_1 = a_2^T \lambda a_1 = \lambda a_2^T a_1 = 0$$

$$a_2^T a_1 = 0$$

$$L(a_2) = a_2^T \Sigma_X a_2 - \lambda (a_2^T a_2 - 1) - \delta a_2^T a_1$$

$$\frac{d}{da_2} L(a_2) = 2 \Sigma_X a_2 - 2 \lambda a_2 - \delta a_1 = 0$$

mult. a_1^T

$$2 a_1^T \Sigma_X a_2 - 2 \lambda a_1^T a_2 - \delta a_1^T a_1 = 0$$

$$\therefore \delta = 0$$

$$2 \Sigma_X a_2 - 2 \lambda a_2 = 0$$

$$\Sigma_X a_2 = \lambda a_2 \leftarrow \lambda_2 \text{ que nos da } a_2$$

$$V(y_2) = \lambda_2$$

$$tr \Sigma_X = \sum_{i=1}^p \sigma_i^2 = \sum_{i=1}^p V(x_i) \quad Y = AX$$

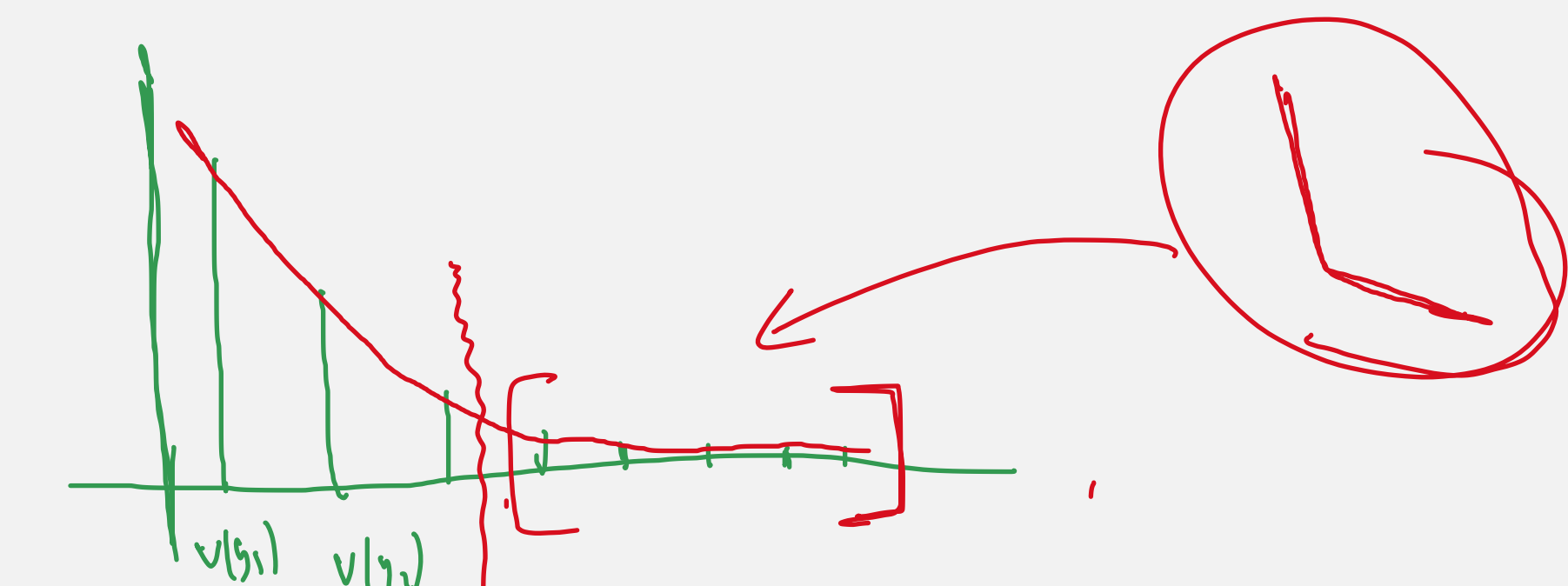
$$\sum_{i=1}^p V(y_i) = tr V(Y) = tr V(AX) = tr (A \Sigma_X A^T) \quad \leftarrow tr(AB) = tr(BA)$$

$$= tr \left(\frac{1}{\sum_{i=1}^p V(y_i)} \Sigma_X \right) = tr(\Sigma_Y) = \sum_{i=1}^p V(x_i)$$

$$V(y_1) \geq V(y_2) \geq V(y_3) \geq \dots$$

* $\frac{V(y_i)}{\sum V(y_i)} \rightarrow$ proporción de aporte a la varianza total

* $\frac{V(y_1) + V(y_2) + V(y_3)}{\sum V(y_i)}$ más cercano a uno mejor



regla del codo